

第一章 矩阵

知识点 矩阵的定义、矩阵的加法、矩阵的数乘、矩阵的乘法、方阵的幂、方阵的多项式、矩阵的转置、对称阵、反对称阵、矩阵的行初等变换求解线性方程组.

课后练习

1. 判断题:

- 1) 任何两个矩阵都可以相加. (X)
 - 2) 任何两个矩阵都可以相乘. (X)
 - 3) 设 $AB=BC$, 则 $A=C$. (X)
 - 4) $(A+B)^2 = A^2 + 2AB + B^2$ 对于任何两个方阵 A 与 B 都成立. (X)
 - 5) $(A+B)(A-B) = A^2 - B^2$ 对于任何两个方阵 A 与 B 都成立. (X)
 - 6) 设 A 是 n 阶方阵, $f(x)$ 和 $g(x)$ 为多项式, 若 $B=f(A), C=g(A)$, 则 $CB=BC$. (✓)
 - 7) 设矩阵 A, B 满足 $AB=O$ (零矩阵) 其中 $A \neq O$, 则 $B=O$. (X)
 - 8) 设矩阵 A 满足 $A^2=O$, 则 $A=O$. (X) $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$
 - 9) 设矩阵 A 满足 $A^2=A$, 则 $A=E$ (单位矩阵) 或 $A=O$. (X) $A = \begin{pmatrix} 1 & a \\ 0 & 0 \end{pmatrix}$
 - 10) 设 A, B 均为 n 阶对称矩阵, 则 AB 是对称矩阵. (X)
 - 11) 设 A 是方阵, 则 $A^T + A$ 是对称矩阵. (✓)
 - 12) 设 A 是方阵, 则 $A^T - A$ 是反对称矩阵. (✓)
2. 设 α 为 3 维列向量, α^T 是 α 的转置, 若 $\alpha\alpha^T = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}$, 则 $\alpha^T\alpha =$ 3 .
设 $\alpha = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ $\alpha\alpha^T = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} (a_1 \ a_2 \ a_3) = \begin{pmatrix} a_1^2 & a_1a_2 & a_1a_3 \\ a_2a_1 & a_2^2 & a_2a_3 \\ a_3a_1 & a_3a_2 & a_3^2 \end{pmatrix}$ $\alpha^T\alpha = a_1^2 + a_2^2 + a_3^2$
3. 已知矩阵 $A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, 计算 $A^2 - BA + AB - B^2 = (A-B)(A+B) =$ _____
 $A^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$ $A^2 - BA + AB - B^2$
 $BA = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 11 & 4 \end{pmatrix}$ $= \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix} - \begin{pmatrix} 5 & 2 \\ 11 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 5 & 8 \end{pmatrix} - \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$
 $AB = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 5 & 8 \end{pmatrix}$ $= \begin{pmatrix} -10 & -10 \\ -17 & -17 \end{pmatrix}$
 $B^2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$

4. 已知阵 $A = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & q \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$, $\xi = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ 满足 $A\xi = \lambda\xi$, 其中 q, λ 为实数, 求 q, λ 的值.

$$A\xi = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & q \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1+q \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow \begin{cases} \lambda = 1 \\ q = 0 \end{cases}$$

5. 已知 $\alpha = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\beta = (1, 2, -4)$, 求 $\alpha\beta$, $\beta\alpha$ 和 $(\alpha\beta)^{100} = \alpha(\beta\alpha)^{99}\beta = (-1)^{99}\alpha\beta = \begin{pmatrix} -1 & -2 & 4 \\ -1 & -2 & 4 \\ -1 & -2 & 4 \end{pmatrix}$

$$\alpha\beta = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} (1 \ 2 \ -4) = \begin{pmatrix} 1 & 2 & -4 \\ 1 & 2 & -4 \\ 1 & 2 & -4 \end{pmatrix}, \quad \beta\alpha = (1 \ 2 \ -4) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -1$$

6. 若 $AB = BA$, 称 A, B 可交换的. 求出所有与 $A = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}$ 可交换的矩阵.

设 $X = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix}$ 与 A 可交换 $X = \begin{pmatrix} 2c+d & 2c \\ c & d \end{pmatrix}$

$$AX = XA \Rightarrow \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}$$

7. 设 $f(x) = x^2 - 2x + 1$, 矩阵 $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, $\begin{pmatrix} x_1+2x_3 & x_2+2x_4 \\ x_1-x_3 & x_2-x_4 \end{pmatrix} = \begin{pmatrix} x_1+x_2 & 2x_1-x_2 \\ x_3+x_4 & 2x_3-x_4 \end{pmatrix}$

$$\Rightarrow \begin{cases} x_1 = 2x_3 + x_4 \\ x_2 = 2x_3 \end{cases}$$

求: (1) $f(A)$, (2) $A^n - 2A^{n-1}$ (n 为大于 1 的自然数).

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix} = 2A$$

$$(1) f(A) = A^2 - 2A + E = 2A - 2A + E = E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(2) n=2, A^2 - 2A = 0, \quad n>2, A^n - 2A^{n-1} = A^{n-2}(A^2 - 2A) = A^{n-2} \cdot 0 = 0$$

8. 已知矩阵 $A = \begin{pmatrix} \lambda & 0 & 0 \\ 1 & \lambda & 0 \\ 0 & 1 & \lambda \end{pmatrix}$, 计算 A^n (n 是正整数).

$$A = B + \lambda E = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$B^2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad B^3 = B^2 \cdot B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B^k = B^3 B^{k-3} = 0 B^{k-3} = 0 \quad (k \geq 3)$$

$$= \begin{pmatrix} \lambda^n & 0 & 0 \\ n\lambda^{n-1} & \lambda^n & 0 \\ \frac{n(n-1)}{2}\lambda^{n-2} & n\lambda^{n-1} & \lambda^n \end{pmatrix}$$

9. 求解下列线性方程组: $A^n = (B + \lambda E)^n = B^n + C_n^1 \lambda B^{n-1} + \dots + C_n^{n-1} \lambda^{n-1} B + \lambda^n E$

$$1) \begin{cases} x_1 - 2x_2 - x_3 + 2x_4 + 4x_5 = 0 \\ 2x_1 - 2x_2 - 3x_3 + 2x_5 = 0 \\ 4x_1 - 2x_2 - 7x_3 - 4x_4 - 2x_5 = 0 \end{cases} = \frac{n(n-1)}{2} \lambda^{n-2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} + n\lambda^{n-1} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \lambda^n \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\bar{A} = \begin{pmatrix} 1 & -2 & -1 & 2 & 4 & 0 \\ 2 & -2 & -3 & 0 & 2 & 0 \\ 4 & -2 & -7 & -4 & -2 & 0 \end{pmatrix} \xrightarrow{\substack{r_2 - 2r_1 \\ r_3 - 4r_1}} \begin{pmatrix} 1 & -2 & -1 & 2 & 4 & 0 \\ 0 & 2 & -1 & -4 & -6 & 0 \\ 0 & 6 & -3 & -12 & -18 & 0 \end{pmatrix}$$

$$\xrightarrow{r_3 - 3r_2} \begin{pmatrix} 1 & -2 & -1 & 2 & 4 & 0 \\ 0 & 2 & -1 & -4 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{r_1 + r_2 \\ r_2 \cdot \frac{1}{2}}} \begin{pmatrix} 1 & 0 & -2 & -2 & -2 & 0 \\ 0 & 1 & -\frac{1}{2} & -2 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

不御 解得 $\begin{cases} x_1 = 2x_3 + 2x_4 + 2x_5 \\ x_2 = \frac{1}{2}x_3 + 2x_4 + 3x_5 \end{cases}$ 其中 x_3, x_4, x_5 为自由未知量

$$2) \begin{cases} x_1 + x_2 + x_3 + x_4 + x_5 = 1, \\ 3x_1 + 2x_2 + x_3 + x_4 - 3x_5 = 0, \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 3, \\ 5x_1 + 4x_2 + 3x_3 + 3x_4 - x_5 = 2. \end{cases}$$

$$\bar{A} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & 1 & -3 & 0 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 5 & 4 & 3 & 3 & -1 & 2 \end{pmatrix} \xrightarrow{\substack{r_2 - 3r_1 \\ r_4 - 5r_1}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -2 & -6 & -3 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & -1 & -2 & -2 & -6 & -3 \end{pmatrix}$$

$$\xrightarrow{r_2 \leftrightarrow r_3, r_3 + r_2, r_4 - r_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1 - r_2} \begin{pmatrix} 1 & 0 & -1 & -1 & -5 & -2 \\ 0 & 1 & 2 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

解得 $\begin{cases} x_1 = x_3 + x_4 + 5x_5 - 2 \\ x_2 = -2x_3 - 2x_4 - 6x_5 + 3 \end{cases}$ 其中 x_3, x_4, x_5 为自由未知量

$$3) \begin{cases} x_1 + x_2 + x_3 + x_4 + x_5 = 1, \\ 3x_1 + 2x_2 + x_3 + x_4 - 3x_5 = 0, \\ x_2 + 2x_3 + 2x_4 + 6x_5 = 0, \\ 5x_1 + 4x_2 + 3x_3 + 3x_4 - x_5 = 2. \end{cases}$$

$$\bar{A} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & 1 & -3 & 0 \\ 0 & 1 & 2 & 2 & 6 & 0 \\ 5 & 4 & 3 & 3 & -1 & 2 \end{pmatrix} \xrightarrow{r_2 - 3r_1, r_4 - 5r_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -2 & -6 & -3 \\ 0 & 1 & 2 & 2 & 6 & 0 \\ 0 & -1 & -2 & -2 & -6 & -3 \end{pmatrix}$$

$$\xrightarrow{\substack{r_3 + r_2 \\ r_4 - r_2}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -2 & -6 & -3 \\ 0 & 0 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

无解组无解

10. 证明题:

1) 设矩阵 $X=(x_1, x_2, \dots, x_n)^T$ 满足 $X^T X=1$. 设 E 是 n 单位矩阵, $H=E-2XX^T$,

证明: H 是对称矩阵, 且 $HH^T=E$.

2) 设 A, B 均为 n 阶反对称矩阵, 则 AB 是反对称矩阵的充要条件为 $AB=-BA$.

3) 设 A 是三阶反对称阵, 证明: 对任意的 $x=(x_1, x_2, x_3)^T$ 都有 $x^T Ax=0$.

$$1) \quad \underline{H^T} = (E - 2XX^T)^T = E^T - 2(XX^T)^T = E - 2XX^T = \underline{H}$$

$$\begin{aligned} HH^T &= (E - 2XX^T)(E - 2XX^T) = E - 2XX^T - 2XX^T + 4(XX^T)(XX^T) \\ &= E - 4XX^T + 4X(\underbrace{X^T X})X^T \\ &= E - 4XX^T + 4XX^T = E \end{aligned}$$

$$4) \quad AB \text{ 为反对称矩阵} \Leftrightarrow (AB)^T = -AB$$

$$\Leftrightarrow B^T A^T = (-B)(-A) = BA = -AB$$

3) $\Rightarrow x^T Ax$ 为一个数, 设为 a

$$\Rightarrow x^T Ax = (x^T Ax)^T = x^T A^T (x^T)^T = x^T (-A)x = -x^T Ax$$

$$\Rightarrow a = -a \Rightarrow a = 0, \quad \forall x \quad x^T Ax = 0.$$